CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 18: Tangents and Intersecting Chords

Every formula and result of the chapter on one sheet

1	Tangent and radius	Perpendicular at the point of contact
2	Tangents from an external point	$PA = PB$
3	Length of a tangent	$\sqrt{d^2 - r^2}$
4	Angle between two tangents	$\angle APB + \angle AOB = 180^\circ$
5	Circles touching externally	Distance between centres $= R + r$
6	Circles touching internally	Distance between centres $= R - r$
7	Intersecting chords (inside or outside)	$PA \times PB = PC \times PD$
8	Tangent and secant	$PT^2 = PA \times PB$
9	Alternate segment theorem	Angle between tangent and chord $=$ angle in the alternate segment
10	Triangle with an incircle	Perimeter $= 2\times$ (sum of the tangent lengths from the three vertices)
11	Quadrilateral circumscribing a circle	$AB + CD = AD + BC$